

Supplemental Material: Strong Optical Disorder Erases Images but Preserves Local Change Observability

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This Supplemental Material is an audit trail for the Letter. It contains (S1) the general Chernoff/Kullback–Leibler jet derivation and its equality cases; (S2) the JET_TEST design with all seventeen frozen checks and the raw exponent tables; (S3) the full-scramble Gaussian derivation, exact generator, rank-one result, and detection duel; (S4) the sealed thin-screen protocol with bars D0–D6, the fixed/fresh comparator, the D3/D5 boundary failures, and the chunking audit; and (S5) the thin-screen covariance aperture and a minimal prior-art fence. All numerical values are read from the committed frozen artifacts JET_TEST (1bf29f1), SCRAMBLE_EXT (ed7a1e0), and SEALED_DET/CONFIRMATORY (b37c841); no new data are generated here.

GENERAL QUOTIENT-JET DERIVATION AND EQUALITY CASES

Let $P_{\varepsilon, \eta}$ be a dominated statistical model in which ε is a scalar physical-change amplitude along a scene direction δ and η is a nuisance. Define the profiled Chernoff distance

$$C_*(\varepsilon) = \inf_{\eta'} \max_{0 \leq s \leq 1} \left[-\log \int p_{\varepsilon, \eta}^s p_{0, \eta'}^{1-s} \right], \quad (\text{S1})$$

and assume the infimum is attained locally with

$$C_*(\varepsilon) = c_m \varepsilon^{2m} + o(\varepsilon^{2m}), \quad c_m > 0. \quad (\text{S2})$$

Theorem A (information-jet detection law). For T independent observations with equal priors,

$$\lim_{T \rightarrow \infty} -\frac{1}{T} \log P_e^*(T, \varepsilon) = C_*(\varepsilon), \quad (\text{S3})$$

so in the joint small-change/large-sample regime with target error exponent h ,

$$T_{\text{req}}(\varepsilon) = \frac{h}{c_m \varepsilon^{2m}} [1 + o(1)]. \quad (\text{S4})$$

For quickest detection at false-alarm level α , if the profiled per-bank KL distance is $D_*(\varepsilon) = k_m \varepsilon^{2m} + o(\varepsilon^{2m})$, then Lorden/CUSUM asymptotics give the mean delay $E_{\text{delay}} = |\log \alpha| / (k_m \varepsilon^{2m}) [1 + o(1)]$. The exponent m is invariant because reparameterizing η leaves the nuisance orbit unchanged, a gain gauge absorbed into η is quotiented automatically, no statistic or Fisher eigenbasis is selected, and physical covariation changes c_m but not the definition of m . The contact order m is robust to an unknown nuisance η : for a truly unknown η the fixed-error *constant* is a composite-hypothesis quantity (a least-favorable-nuisance / GLRT exponent) for which the profiled Chernoff distance is a bound, but the exponent m itself is unchanged.

Gaussian specialization. Let $P_\varepsilon = \mathcal{N}(0, \Sigma_\varepsilon)$ with $\Sigma_\varepsilon = \Sigma_0 + z(\varepsilon)H$, $\Sigma_0 \succ 0$, and put $A_\varepsilon = z(\varepsilon)K$, $K = \Sigma_0^{-1/2} H \Sigma_0^{-1/2}$. The exact divergences are

$$D(P_\varepsilon \| P_0) = \frac{1}{2} [\text{tr } A_\varepsilon - \log \det(I + A_\varepsilon)] = \frac{z^2}{4} \text{tr}(K^2) - \frac{z^3}{6} \text{tr}(K^3) + O(z^4), \quad (\text{S5})$$

$$B(P_\varepsilon, P_0) = \frac{1}{2} \log \det \left(I + \frac{A_\varepsilon}{2} \right) - \frac{1}{4} \log \det(I + A_\varepsilon) = \frac{z^2}{16} \text{tr}(K^2) + O(z^3). \quad (\text{S6})$$

For nearby regular distributions the optimizing Chernoff parameter tends to $s = \frac{1}{2}$, so Eq. (S6) is the leading Chernoff coefficient.

Equality and boundary cases. With $z(\varepsilon) = 2c\varepsilon + d\varepsilon^2$, $c = x^\top G\delta$, $d = \delta^\top G\delta > 0$: (i) fast class, $c \neq 0$: $D = c^2\|K\|_F^2\varepsilon^2 + o(\varepsilon^2)$, $m = 1$; (ii) curvature class, $c = 0$, $d > 0$: $D = \frac{1}{4}d^2\|K\|_F^2\varepsilon^4 + o(\varepsilon^4)$, $m = 2$; (iii) true local blindness only if $d = 0$ (a nullspace of G at coarse grain) or the covariance signature is entirely nuisance; (iv) global finite-change cancellation for signed changes at $\varepsilon_* = -2c/d$, so the theorem is local; on the monotone cone ($x \geq 0$, $\delta \geq 0$, G entrywise nonnegative) $c, d \geq 0$ and every $\varepsilon > 0$ gives $\Delta Q > 0$; (v) amplitude gauge: if $\Sigma = R + aQ(x)H$ then $\partial_a\Sigma = QH$ and $\partial_Q\Sigma = aH$ are collinear, so zero-lag covariance carries zero efficient information for separating Q from a . Under nuisance profiling with $\langle A, B \rangle_0 = \frac{1}{2}\text{tr}(\Sigma_0^{-1}A\Sigma_0^{-1}B)$ and $H_\perp = (I - \Pi_\eta)H$, Eqs. (S5)–(S6) hold with $\|K_\perp\|_F$, $K_\perp = \Sigma_0^{-1/2}H_\perp\Sigma_0^{-1/2}$; if $H_\perp = 0$ the change is unidentifiable from that ledger.

JET_TEST DESIGN AND THE SEVENTEEN FROZEN CHECKS

The jet test uses the exact scrambling generator (Sec. S3) on a 32×32 grid ($N = 1024$), code band $P_B = 3$, grain band $K = 13$, $M = 36$ codes, and 10^4 photons per bucket, with $\|K\|_F^2 = 3.46 \times 10^{-4}$. Three scene directions are constructed: a generic beyond-band direction δ_g with $c = x^\top G\delta_g \neq 0$; an exactly G -orthogonal direction δ_o with $x^\top G\delta_o = 0$; and a small-eigenvalue direction δ_n . A mixed direction δ_{mix} with a small nonzero c probes the crossover.

Table S1 lists the raw contact orders and coefficient ratios. The exact-KL log-log slopes are 2.038 (δ_g) and 4.000 (δ_o, δ_n), matching the predicted integer orders; the measured leading coefficients agree with Eqs. (S5)–(S6) to 1.07% (δ_g) and 0.00% (δ_o). The Monte Carlo single-look $d' \propto \varepsilon^m$ exponents are 0.95 and 2.05; the CUSUM (ARL₀ = 500, thresholds set by the ARL₀ approximation of [12]) delay slopes are -2.16 and -3.92 . These two CUSUM slopes are two-point confirmations of the densely sampled (12-point, bank A) exact-KL contact orders, illustrative rather than independent fits. The mixed direction crosses over at $\varepsilon_{\text{cross}}^{\text{pred}} = 9.995 \times 10^{-3}$ versus empirical 9.997×10^{-3} (ratio 1.0003).

TABLE S1. Raw jet exponents (committed JET_TEST, 1bf29f1). $c = x^\top G\delta$, $d = \delta^\top G\delta$; KL slope is the log-log regression of the exact Gaussian KL over an amplitude decade; coef. ratio is measured/predicted leading coefficient.

Direction	c	d	m_{pred}	KL slope	Bhatt. slope	coef. ratio
δ_g (generic)	29.78	164.26	1	2.0381	2.0374	1.0107
δ_o (orthogonal)	2.0×10^{-15}	137.43	2	3.9999	3.9999	1.0000
δ_n (small eig.)	-3.1×10^{-6}	44.19	2	4.0000	4.0000	0.9999

TABLE S2. Monte Carlo and sequential-detection slopes, and the nuisance quotient (committed JET_TEST). I_θ is the per-bank efficient information for log fluctuation scale; “naive” omits nuisance profiling.

Quantity	δ_g / value	δ_o / value
MC d' exponent m	0.955	2.046
CUSUM delay slope	-2.158	-3.916
Amplitude gauge, single zero lag: I_θ^{naive}		0.614
Amplitude gauge, single zero lag: I_θ^{eff}		0.000
Amplitude gauge, proportional lags: I_θ^{eff}		0.000
Second grain kernel: fraction recovered		0.018
Amplitude anchor: fraction recovered		0.200

The signed iso- Q cancellation crosses at $\varepsilon_* = 0.363$ with AUC = 0.481 (chance), while the two sides remain visible (AUC = 0.026 and 1.000), confirming locality. The monotone cone gives $\Delta Q > 0$ for all seven tested amplitudes (ΔQ from 1.59 at $\varepsilon = 0.005$ to 156.0 at $\varepsilon = 0.4$), with the kernel entrywise nonnegative. A scan of 400 in-aperture directions gives $d_{\text{min}} = \delta^\top G\delta = 0.254$ (median 0.265), so the blind fraction is exactly 0. All seventeen frozen checks pass and none of the four kill conditions fire (Table S3; verdict MOONSHOT_SURVIVES).

TABLE S3. All seventeen predeclared JET_TEST checks (committed JET_TEST, 1bf29f1); every entry is **true**, and each of the four kill conditions is **false**.

Check	Status
1. KL contact order generic = 2	pass
2. KL contact order orthogonal = 4	pass
3. orders integer	pass
4. generic coefficient within 10%	pass
5. orthogonal coefficient within 10%	pass
6. crossover within 25%	pass
7. delay slope generic near -2	pass
8. delay slope orthogonal near -4	pass
9. MC generic m near 1	pass
10. MC orthogonal m near 2	pass
11. amplitude gauge collapse to zero	pass
12. collapse persists across lags	pass
13. information returns with anchor	pass
14. AUC returns to $\frac{1}{2}$ at ε_*	pass
15. locally visible either side of ε_*	pass
16. cone $\Delta Q > 0$ for all tested	pass
17. no broad in-aperture blind set	pass
Kills (non-integer order; ε^{-4} class vanished;	all
info survives collinear amplitude; broad blind set)	false

FULL-SCRAMBLE GAUSSIAN DERIVATION, GENERATOR, RANK-ONE RESULT, AND DUEL

A band-limited modulator imprints a real, nonnegative code p_i (Fourier support B_p) on the illumination; a dynamic strongly scattering medium with state T_t produces the speckle field $E_{i,t}(r) = \sum_a T_t(r, a) A_i(a)$, and a single-pixel bucket collects $b_{i,t} = \sum_r |E_{i,t}(r)|^2 x(r) + \text{shot}$. Fully developed speckle means T_t has zero-mean circular-complex-Gaussian entries, so $E_{i,t}$ is a zero-mean circular-complex-Gaussian field with mutual coherence $J_{ij}(r, r') = \langle E_{i,t}(r) E_{j,t}(r')^* \rangle$. With medium statistics $\langle T_t(r, a) T_t(r', b)^* \rangle = C(r, r') \Gamma(a, b)$ and full scrambling $\Gamma(a, b) = \tau(a) \delta_{ab}$, the coherence factorizes,

$$J_{ij}(r, r') = C(r - r') O_{ij}, \quad O_{ij} = I_0 \sum_a \tau(a) p_i(a) p_j(a). \quad (\text{S7})$$

The joint field is exactly generated by $E_t = B(Z_t O^{1/2})$ with Z_t i.i.d. unit circular-Gaussian, $O^{1/2}$ the code-covariance root, and $BB^\dagger = C$; this Kronecker generator drives the simulation with no approximation.

Mean wall. The mean intensity is $\mu_i(r) = J_{ii}(r, r) = C(0) O_{ii}$, independent of r , so $\mathbb{E}[b_i] = C(0) O_{ii} \hat{x}(0)$: the first-moment channel measures only the scene DC term, with spatial band $B_{\text{mean}} = \{0\}$. Every h with $\hat{h}(0) = 0$ is annihilated. Numerically the mean envelope coefficient of variation is 0.0156 (Monte Carlo floor 0.0158), and the mean response to a beyond-band, DC-free change has relative magnitude 6.56×10^{-19} .

Covariance and rank collapse. By the Siegert relation $\text{Cov}_t[S_{i,t}(r), S_{j,t}(r')] = |J_{ij}(r, r')|^2$, so with Eq. (S7) the same-epoch bucket covariance collapses to

$$\text{Cov}_t[b_{i,t}, b_{j,t}] = O_{ij}^2 Q(x), \quad Q(x) = \sum_{r, r'} x(r) x(r') |C(r - r')|^2 = x^\top G x. \quad (\text{S8})$$

In Fourier, $Q(x) = \sum_k \hat{G}(k) |\hat{x}(k)|^2$ with $\hat{G} = |\widehat{C}|^2$ supported on $|k| \leq 2k_{\text{grain}}$. The measured aperture is full ($\hat{G}$ nonzero on all 1024/1024 bins, $C_0 = 0.517$), yet the scene enters only through the scalar Q , so the covariance Fisher information is rank one, $I_{xx}^{\text{cov}} = 4\kappa(Gx)(Gx)^\top$. Empirically $\hat{Q}$ from $\text{Var}[b_i]/O_{ii}^2$ is 190.9 against the theoretical $Q(x) = 187.4$ (ratio 1.019), and 99.99% of the covariance matrix energy lies in the single direction $O \circ O$. The change functional obeys $\Delta Q = 2x^\top G \delta + \delta^\top G \delta$; the predicted $\Delta Q = 0.34803$ matches the measured 0.34803 to relative error 2×10^{-15} , split into cross term 0.0031 and quadratic term $0.3449 > 0$ for the orthogonal direction.

Detection duel. Table S4 reports the H0-vs-H1 duel (efficient covariance score) at the complete-scramble endpoint. The mean detector is at chance for both a coherent and an orthogonal beyond-band 5% change (AUC = 0.502 and 0.505), while the covariance detector reaches $d' = 4.10$, AUC = 0.997 for the generic 5% change at 8192 banks. The worst-case orthogonal direction obeys the predicted $d' \sim \varepsilon^2 \sqrt{T}$ law and is never zero; its slower rate ($a = 0.0050$ per $\sqrt{T}$ vs 0.0459 for the coherent change) is the curvature-class ε^{-4} price, not a blind direction.

TABLE S4. Complete-scramble detection (committed **SCRAMBLE_EXT**, **ed7a1e0**). d' and AUC at listed bank budgets T_{eff} ; the mean detector uses 512 banks. “coherent” is a generic direction ($x^\top G \delta \neq 0$); “orthogonal” is G -orthogonal.

Channel / change	T_{eff}	d'	AUC
Mean, coherent 5%	512	0.030	0.502
Mean, orthogonal 5%	512	0.015	0.505
Covariance, coherent 5%	512	0.99	0.762
Covariance, coherent 5%	2048	2.22	0.943
Covariance, coherent 5%	8192	4.10	0.997
Covariance, coherent 2%	8192	1.60	0.887
Covariance, orthogonal 5%	8192	0.36	0.601
Covariance, orthogonal 5%	65536	1.23	0.809

SEALED THIN-SCREEN PROTOCOL, BARS D0–D6, COMPARATOR, AND AUDITS

The sealed sentinel uses a true zero-dimensional bucket record on a 64×64 scene ($N = 4096$), modulator band $k_p = 5$, $M = 128$ signed effective codes realized by 256 complementary nonnegative exposures per bank at a 20 kHz pattern rate (12.8 ms per bank; cap $T_{\text{eff}} = 4096$ banks = 52.4 s), 10^4 detected photoelectrons per physical bucket, and a medium quasi-static within a bank and independent between banks. Thresholds were committed before confirmatory generation; the run used 12 confirmatory banks and 2.76 of a 6.0 GPU-hour ceiling. Table S5 summarizes the seven bars.

TABLE S5. Sealed detector bars D0–D6 (committed **SEALED_DET/CONFIRMATORY**, **b37c841**; analysis freeze **5910277**). “passed” is the frozen bar verdict; D3 and D5 did not pass their strong forms but did not trigger a kill, defining the honest boundary (Sec. S4).

Bar	Key measured endpoint	Verdict
D0 mechanism/engine	mean deriv. 5.7×10^{-16} ; two-engine 1.7%; shot ledger 1.0%	pass
D1 calibration	$n = 108$ cells, median $d'_{\text{emp}}/d'_{\text{an}} = 0.998$, all in $[0.91, 1.10]$	pass
D2 capability	2% in 77/81 cells, best 453 banks, MC power LCB 0.990	pass
D3 specificity	TPR 0.988, bal. acc. 0.916; medium FA 0.096/0.084 > 0.05	boundary
D4 fresh comparator	fixed vs fresh latency 459 vs 1013 banks (retain fixed)	pass
D5 mismatch	AUC 1.000; $\leq 10\%$ rotation passes; 20%/slope–1 fail	boundary
D6 online feasibility	0.0203 ms (0.16% of bank), 0.26 MB	pass

D2 capability. The 2% beyond-band change is analytically detectable within the 4096-bank cap in 77/81 predeclared cells; the best cell (unit static factor, k^{-2} spectrum, $k_w/k_p = 1$, $1.25k_p$ claim shell) requires 453 banks and the pooled Monte Carlo power lower bound is 0.990 ($n = 13500$). The 5% change clears all 81 energy-spread cells and 75/81 worst-mode cells; the 1% best cell clears $d' \geq 5$ by 1811 banks; the 0.5% edge audit reaches $d'_{\text{max}} = 3.76$.

D3 specificity (strong claim killed, power intact). At the threshold calibrated to 1% H0 false alarm ($t = 0.051$), the target true-positive rate is 0.988, in-band false alarms are 0.020, and four-class balanced accuracy is 0.916; the beyond score stays centered on non-targets ($|d'| \leq 0.019$) and the attribution-simplex maximum off-diagonal canonical correlation is 0.039. However, medium-amplitude and medium-timescale events exceed the frozen 0.05 false-alarm bar at 0.096 and 0.084, and the weakest class-specific separation is 1.79 rather than 5 (the intended per-class separations are $[1.79, 43.06, 4.78]$). The result is a power-certified beyond-band alarm with measured attribution leakage, not a fully specific sentinel; no “specific”, “zero-false-alarm”, or “medium-immune” wording is used.

D4 fixed-versus-fresh. On identical photons, exposures, duration, anomaly bank, and law knowledge, the repeated (fixed) code bank retains a latency advantage: best cell 459 (fixed) versus 1013 (fresh) banks, mid cell 1271 versus 1488, floor cell 1598 versus 3334; the frozen branch retains the fixed-code concentration.

D5 mismatch (robustness frozen to $\leq 10\%$ rotation). AUC is 1.000 on every tested axis and latency inflation is bounded, but false-alarm calibration fails outside the primary domain: at aperture-preserving basis rotation 10% the non-target false alarm is 0.032 (latency inflation 19%, pass), whereas at 20% rotation it is 0.052 and at spectral-slope mismatch -1 it is 0.076 (fail). Robustness is therefore frozen to the $\leq 10\%$ rotation regime; the kill-tree verdict is “narrow the physical domain or kill robustness wording”, and the broader wording is withdrawn.

Integrity audits (verbatim). *Record generator.* After protocol freeze and before unblinding, the record generator was refactored solely to stream Monte Carlo records in GPU-memory chunks; an audited diff found no changes to hypotheses, thresholds, bank counts, scenes, estimators, or endpoints, although altered random-number consumption produced a different, still-blinded realization set. All reported decisions apply the frozen analysis unchanged to that disclosed chunked stream. *Shot model.* A mandated independent-engine check exposed a signed-code shot-noise error that had inflated covariance Fisher information by approximately $1.9\times$; replacing the signed difference Px with the physical complementary-exposure throughput $|P|x$ reduced the frozen $P1$ witness coverage from 0.050 to 0.033 and eliminated the apparent natural-scene living pocket, so all values reported here come from the corrected artifacts and every prior kill only strengthened.

THIN-SCREEN COVARIANCE APERTURE AND MINIMAL PRIOR-ART FENCE

Aperture interpolation. The thin multiplicative screen is the weakest complex medium: a single random phase layer w_t . Its bucket covariance is $C(x) = \Phi^2 P \text{diag}(x) K_w \text{diag}(x) P^\top$, whose informative support is the convolution aperture $B_w \oplus B_p$ and whose Fisher structure is rich—many recoverable scene modes. As the medium approaches complete scrambling, the input coupling Γ loses angular memory (rank $r_{\text{ME}} \rightarrow 1$) and the covariance factorizes as in Eq. (S7), collapsing all $M(M+1)/2$ entries to the single scalar $Q(x)$. For a residual memory effect of rank r_{ME} the covariance becomes a rank- $O(r_{\text{ME}}^2)$ combination of quadratic scene functionals, so the recoverable imaging aperture interpolates monotonically from the rich thin-screen aperture to the rank-one scalar. *Beyond-band imaging survives exactly to the extent the medium retains memory-effect modes; beyond-band detection survives complete scrambling,* because $\delta^\top G \delta > 0$ for every nonzero δ . This is the precise, physically correct boundary between the two disorder endpoints of the Letter.

Minimal prior-art fence. The result is distinct from the head-on prior art. Memory-effect imaging through opaque scattering layers [6–8] *reconstructs* a hidden object while residual angular correlations survive; we operate at the endpoint where reconstruction rank is one and only the observability order survives. Blind structured-illumination and speckle-superresolution imaging [1, 2] and photoacoustic speckle-fluctuation imaging [3] likewise *reconstruct* by exploiting residual correlations; our contribution is the opposite regime. Quantum-inspired superresolution [4] moves Fisher information between optical *measurements* of the same scene, and quantum change detection [9] optimizes quantum-limited discrimination of a *known* object change; the Rank-Jet Separation instead moves contact order between *statistical layers* of one scalar classical record and classifies a *nuisance-profiled* contact order, with no change of hardware. Speckle-decorrelation and coda-wave change detection [5] and SAR coherent change detection [10] sense inter-pass decorrelation of the *medium*; here an exact first-moment invariance wall makes the medium a transducer for *scene* change while a profiled quotient rejects medium-law events. Mesoscopic universal conductance fluctuations [11] are the disorder-statistics ancestor of the covariance channel but carry no detection-order classification. No claim of prior-art equivalence is made, and the novelty is the nuisance-profiled statistical jet order as a physical observability class for disordered sensing.

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- [1] E. Mudry *et al.*, Nat. Photonics **6**, 312 (2012).
 - [2] J. Idier *et al.*, IEEE Trans. Comput. Imaging **4**, 87 (2018).
 - [3] T. Chaigne *et al.*, Optica **3**, 54 (2016).
 - [4] M. Tsang, R. Nair, and X.-M. Lu, Phys. Rev. X **6**, 031033 (2016).
 - [5] R. Snieder, A. Grêt, H. Douma, and J. Scales, Science **295**, 2253 (2002).
 - [6] J. Bertolotti, E. G. van Putten, C. Blum, A. Lagendijk, W. L. Vos, and A. P. Mosk, Nature **491**, 232 (2012).
 - [7] O. Katz, P. Heidmann, M. Fink, and S. Gigan, Nat. Photonics **8**, 784 (2014).
 - [8] I. Freund, M. Rosenbluh, and S. Feng, Phys. Rev. Lett. **61**, 2328 (1988).

- [9] M. R. Grace and S. Guha, Phys. Rev. Lett. **129**, 180502 (2022).
- [10] M. Preiss and N. J. S. Stacy, *Coherent Change Detection: Theoretical Description and Experimental Results*, DSTO-TR-1851 (2006).
- [11] P. A. Lee and A. D. Stone, Phys. Rev. Lett. **55**, 1622 (1985).
- [12] D. Siegmund, *Sequential Analysis: Tests and Confidence Intervals* (Springer, New York, 1985).